ME5311 Project Report: Data-Driven Analysis of a 2D Field Dataset

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This report studies a time-varying 2D field dataset using POD and Fourier spectra to summarize dominant structures, spatial scales, periodic signatures, and anisotropy. Most variation is captured by a limited number of POD modes: 95% cumulative energy is reached at $r = 32$, and a rank-4 reconstruction reproduces the main spatial patterns. The isotropic spectrum $E(k)$ decays with increasing k, so energy is concentrated at large spatial scales. In the time-averaged Fourier spectrum, a distinct peak appears at $k_x = -0.0625$ and $k_y = 0$, corresponding to a wavelength of about 16 grid units. Directional spectral cuts show stronger variation along the x direction than along y, indicating clear anisotropy.

POD | Fourier spectrum | anisotropy | characteristic length scale | reduced-order modeling

Introduction

This report presents a concise yet structured analysis of a time-resolved 2D field dataset. Dominant spatial patterns are identified, and the number of components required to represent most of the variability is evaluated. The distribution of energy across spatial scales is examined to clarify the relative roles of large- and small-scale structures. Evidence for a distinct periodic spatial signature is assessed through the presence of a clear spectral peak and its corresponding characteristic wavelength. Directional behavior is further compared between the x and y directions to determine whether anisotropy exists. Finally, the spatial findings are connected to the dominant time-scale content associated with the leading patterns (1, 2).

Methods

Data Representation. The dataset is a time-resolved two-dimensional field sampled on a fixed grid. Each frame is treated as one snapshot in time (3). For matrix-based analysis, every snapshot is reshaped into a column vector and stacked to form a snapshot matrix

$$X \in \mathbb{R}^{N \times T},$$

where $N = N_x N_y n_c$, with $N_x = N_y = 64$ and $n_c = 2$ components, and T is the number of time frames. This representation allows dominant spatial patterns and their time variation to be extracted in a consistent and compact form.

POD and Reconstruction. Proper orthogonal decomposition (POD) is computed via singular value decomposition (4),

$$X = U \Sigma V^{\top}.$$

The spatial modes are given by the columns of U and are ordered by decreasing importance. Mode importance is quantified using the normalized energy and cumulative energy,

$$E_r = \frac{\sigma_r^2}{\sum_j \sigma_j^2}, \qquad E_{\mathrm{cum}}(r) = \sum_{i=1}^{r} E_i,$$

which directly indicates how many modes are needed to represent a target fraction of the dataset variance. To assess how well a compact representation preserves the field, a rank-r approximation is used for reconstruction and compared against the original snapshots (5). Reconstruction quality is reported using global error metrics, including the relative Frobenius error and RMSE, which summarize overall mismatch across the full space–time dataset.

Spatial Spectra and Anisotropy. Spatial-scale behavior is examined in Fourier space (6, 7). A 2D Fourier transform is applied to each snapshot, and the time-averaged 2D energy spectrum

$$E(k_x, k_y)$$

is computed from the squared magnitude of the Fourier coefficients and averaged over time. This spectrum reveals where energy concentrates in wavenumber space and provides a direct way to detect dominant length scales. An isotropic spectrum $E(k)$ is obtained by ring-averaging in wavenumber space, which summarizes the overall scale-dependent decay of energy without emphasizing any particular direction. Directional behavior is evaluated using the DC-axis cuts

$$E(k_x, 0), \qquad E(0, k_y),$$

and consistent differences between these cuts are interpreted as evidence of anisotropy between the x and y directions (8–10). These axis cuts are extracted from the time-averaged 2D spectrum and interpreted together with finite-resolution effects near the high-k end.

Periodic Signature and Time Scales. A periodic spatial component is taken as present when $E(k_x, k_y)$ contains a peak clearly above the surrounding background. The peak location gives the dominant wavenumber, and with wavenumber reported in cycles per grid unit, the associated characteristic wavelength is estimated by

$$\lambda^* \approx \frac{1}{|k^*|},$$

giving λ^* in grid units. Temporal behavior is characterized using the power spectra of the leading POD coefficients $a_r(t)$ to show whether dominant dynamics are concentrated at low frequencies and to estimate characteristic time scales associated with the main spatial patterns (11).

Results and Discussion

Figure A and Figure B show how much of the dataset is carried by coherent low-rank structure. In Figure A, the first four POD modes already contain 85.4% of the total energy, while 95% cumulative energy requires $r = 32$. This split separates core structure from a weaker but still noticeable modal tail. The dominant spatial organization is concentrated in a very small set of modes, but additional weaker modes are still needed to represent the remaining variability. Figure B illustrates this distribution. The leading modes show coherent wave-like and banded shapes, so the dominant variance is structured rather than random texture. A rank-four reconstruction reproduces this large-scale organization yet remains imperfect. The relative Frobenius error is 0.283 and the RMSE is 0.300. Together, these errors show that the main geometry and orientation are reproduced, while higher-order content in the energetic tail is smoothed.

Figure C and Figure F clarify scale content and isolate a periodic spatial component. Figure C shows that the isotropic spectrum decays as the wavenumber magnitude increases, so energy is dominated by large spatial scales and diminishes toward smaller scales. At high wavenumbers (around $k \approx 0.35$ and above), the spectrum approaches a numerical noise floor associated with the finite grid limit near Nyquist. This trend matches the low-rank behavior seen in Figure A. When energy is concentrated at low wavenumbers, the field can be represented by a small number of coherent modes. Figure F goes beyond the general decay and shows a distinct peak at $k_x = -0.0625$ and $k_y = 0$. The associated dominant wavelength is about 16 grid units. The peak-to-background ratio is about 22.1, so the periodic component is not a minor broadband fluctuation. Because the peak sits at $k_y = 0$, the periodic content is mainly aligned with the x direction, matching the directional imbalance in the axis cuts.

Figure E and Figure D connect directional bias with time-scale content. In Figure E, the spectrum along the x direction is substantially stronger than along y. The integrated anisotropy ratio is 5.80, and the peak-based anisotropy ratio is 7.74. The larger peak-based value means the most energetic structures are more anisotropic than the background distribution. The sharp high-k roll-off in these axis cuts comes from finite spectral resolution and the same noise-floor behavior seen in Figure C. This pattern matches Figure F, where the dominant peak lies on $k_y = 0$ and indicates a preferred spectral direction. Figure D then shows that the leading POD coefficients are dominated by low-frequency peaks, with characteristic periods ranging from 375 to 3000 time units. These spectra indicate that the coherent structures identified in Figure A and Figure B are not rapidly fluctuating. They evolve on relatively long time scales, in line with the large-scale dominance in Figure C and the persistent peak in Figure F.

Conclusion

The dataset is dominated by a small number of coherent POD modes, and a low-rank reconstruction already captures the main spatial organization. Energy is concentrated at large spatial scales, and the Fourier spectrum reveals a strong periodic component with a dominant wavelength of about 16 grid units. Clear anisotropy is present, with substantially higher spectral energy along the x direction than along y. The dominant POD coefficients are low-frequency, indicating that the main structures evolve on relatively long time scales.

AI Declaration. AI tools, including GPT-5.2, were used for language polishing and minor code modifications.

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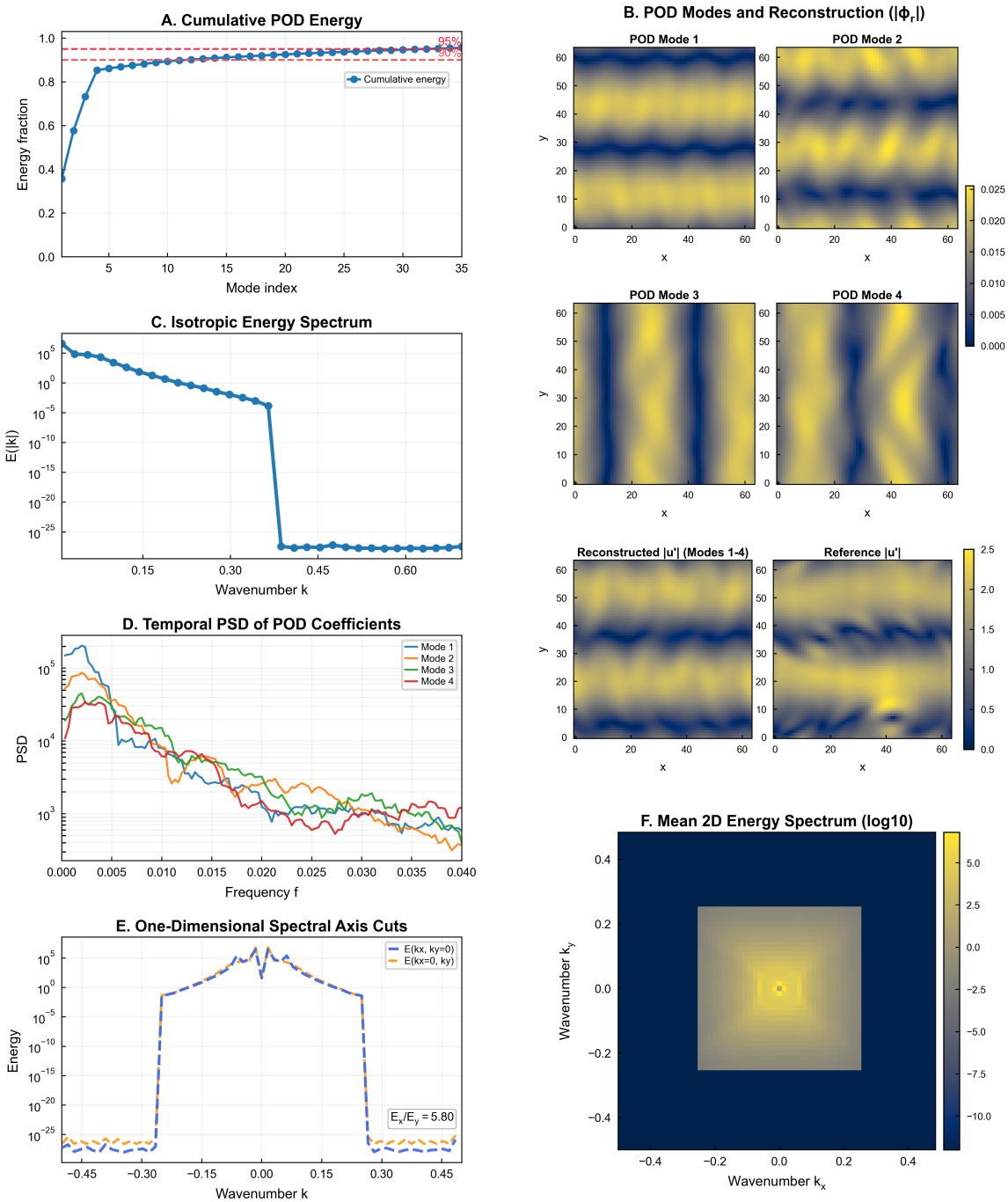

Fig. 1. POD and Spectral Analysis of a Time-Varying 2D Field

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